

Sexual Reproduction in Flowering Plants

Chapter 1 · All questions with full solutions



Part 3 · Questions and Solutions

Q

1

Sites of gametophyte development in a flower



Name the parts of an angiosperm flower in which development of male and female gametophyte take place.

Conclusion

The male gametophyte, or pollen grain, develops inside the pollen chamber of the anther.

The female gametophyte, also called the embryo sac, develops inside the nucellus of the ovule from the functional megaspore.

Q

2

Microsporogenesis vs megasporogenesis



Differentiate between microsporogenesis and megasporogenesis. Which type of cell division occurs during these events? Name the structures formed at the end of these two events.

Microsporogenesis

1. It is the process of the formation of microspore tetrads from a microspore mother cell through meiosis.

2. It occurs inside the pollen sac of the anther.

Megasporogenesis

It is the process of the formation of the four megaspores from a megaspore mother cell in the region of the nucellus through meiosis

It occurs inside the ovule.

Conclusion

Both events, microsporogenesis and megasporogenesis, involve meiosis (reduction division).

This forms haploid gametes from the microspore and megaspore mother cells.

Microsporogenesis forms haploid microspores from a diploid microspore mother cell.

Megasporogenesis forms haploid megaspores from a diploid megaspore mother cell.

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3

Developmental sequence of pollen formation

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Arrange the following terms in the correct developmental sequence: Pollen grain, sporogenous tissue, microspore tetrad, pollen mother cell, male gametes.

Given

The correct development sequence is as follows:

Sporogenous tissue - pollen mother cell - microspore tetrad - Pollen grain - male gamete

Substitute

During the development of the microsporangium, each cell of the sporogenous tissue becomes a pollen mother cell.

It undergoes meiosis (microsporogenesis) to form a microspore tetrad of four haploid microspores.

As the anther matures, these microspores separate and develop into pollen grains.

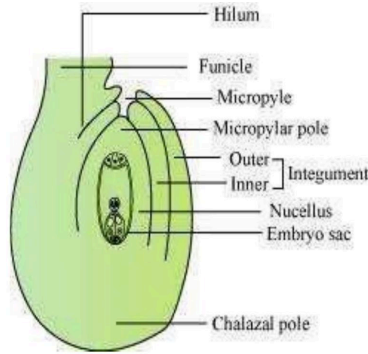
Conclusion

The pollen grains then mature and produce male gametes.

Q

4

Structure of
a typical
angiosperm
ovule



With a neat, labelled diagram, describe the parts of a typical angiosperm ovule.

Given

An ovule is the female megasporangium.

Megaspores form inside it.

Parts of the ovule: The various parts of an ovule are -

- Funiculus - A stalk-like structure that attaches the ovule to the placenta of the ovary.
- Hilum - The point where the body of the ovule joins the funiculus.
- Integuments - The outer layers around the ovule.

They protect the developing embryo.

- Micropyle - A narrow pore formed by the projection of the integuments.

The pollen tube enters the ovule through this pore at fertilization.

- Nucellus - A mass of parenchymatous tissue enclosed by the integuments from the outside.

It nourishes the developing embryo, and the embryo sac lies inside it.

- Chalazal - The swollen basal part of the nucellus from which the integuments arise.

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5

Monosporic
development
of female
gametophyte



What is meant by monosporic development of female gametophyte?

Monosporic development

The female gametophyte, or embryo sac, develops from a single functional megaspore.

This is called monosporic development of the female gametophyte.

Conclusion

In most flowering plants, a single megaspore mother cell lies at the micropylar pole of the nucellus.

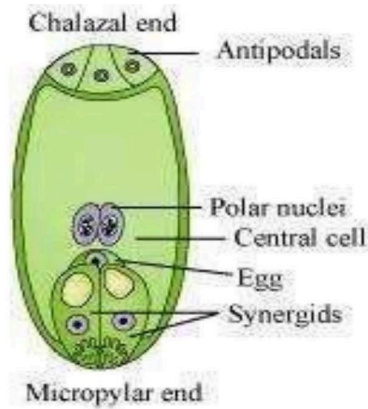
It undergoes meiosis to form four haploid megaspores.

Of these four, only one functional megaspore survives and develops into the female gametophyte, while the remaining three degenerate.

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6

7-celled 8-nucleate female gametophyte



With a neat diagram explain the 7-celled, 8-nucleate nature of the female gametophyte.

Given

The female gametophyte (embryo sac) develops from a single functional megaspore. This megaspore divides mitotically three times in succession to form an eight-nucleate embryo sac.



Substitute

Three mitotic divisions: The first mitotic division forms two nuclei.

One nucleus moves towards the micropylar end, and the other moves towards the chalazal end.

Each of these nuclei then divides again, and divides once more, giving eight nuclei in all - four at the micropylar end and four at the chalazal end.

Egg apparatus and antipodal cells: At the micropylar end, three of the four nuclei become two synergids and one egg cell.

Together these three cells form the egg apparatus.

At the chalazal end, three of the four nuclei become the three antipodal cells.

Polar nuclei: The one remaining nucleus at each end moves towards the centre.

These two nuclei are called the polar nuclei, and they lie together in a large central cell.

Conclusion

Hence, at maturity, the female gametophyte has eight nuclei but appears as a 7-celled structure.

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7

Chasmogamous vs cleistogamous flowers

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What are chasmogamous flowers? Can cross-pollination occur in cleistogamous flowers? Give reasons for your answer.

Given

Plants such as Oxalis and Viola have two types of flowers: chasmogamous and cleistogamous.



Chasmogamous flowers: Chasmogamous flowers have exposed anthers and stigma, like the flowers of most other species.

Cleistogamous flowers: Cleistogamous flowers cannot cross-pollinate.

They never open at all, and their anther and stigma lie close to each other.

So only self-pollination is possible in these flowers.

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Strategies
preventing
self-
pollination



Mention two strategies evolved to prevent self-pollination in flowers.

Conclusion

Self-pollination is the transfer of pollen from the stamen to the pistil of the same flower.

Two strategies have evolved to prevent self-pollination in flowers:

- In some plants, the stigma can stop pollen grains from germinating, so the pollen tube cannot grow.

This genetic mechanism is called self-incompatibility.

It can occur between individuals of the same species or of different species.

It prevents breeding between them.

- In some plants, the gynoecium matures before the androecium, or the androecium matures before the gynoecium.

These are called protogyny and protandry respectively.

Either way, the pollen cannot reach the stigma of the same flower.

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Self-
incompatibili-
ty and seed
formation



What is self-incompatibility? Why does self-pollination not lead to seed formation in self-incompatible species?

Self-incompatibility

Self-incompatibility is a genetic mechanism in angiosperms that prevents self-pollination.

It creates genetic incompatibility between individuals of the same species or between individuals of different species.

Conclusion

Plants with this trait can stop pollen grains from germinating on the stigma, so the pollen tube cannot grow.

This blocks fusion of the gametes and stops the embryo from developing.

As a result, no seed forms.

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Bagging
technique in
plant
breeding



What is bagging technique? How is it useful in a plant breeding programme?

Bagging technique

In crop improvement programmes, artificial hybridization often begins with emasculation - removing the anther from a bisexual flower without affecting the female reproductive part, the pistil.

The emasculated flower is then wrapped in a bag to prevent pollination by unwanted pollen grains.

This process is called bagging.

Conclusion

Bagging is an important part of the plant breeding programme.

It ensures that pollen grains of only desirable plants are used to fertilize the stigma, so that the desired plant variety develops.

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Triple fusion
in the
embryo sac

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What is triple fusion? Where and how does it take place? Name the nuclei involved in triple fusion.

Triple fusion

Triple fusion is the fusion of a male gamete with two polar nuclei inside the embryo sac of the angiosperm.

Substitute

This fusion takes place inside the embryo sac.

Pollen tube growth: When a pollen grain falls on the stigma, it germinates and gives rise to a pollen tube.

The tube passes through the style, enters the ovule, and then enters one of the synergids, where it releases two male gametes.

Syngamy: One of the two male gametes fuses with the nucleus of the egg cell and forms the zygote.

This is called syngamy.

Triple fusion and endosperm formation: The other male gamete fuses with the two polar nuclei present in the central cell, forming a triploid primary endosperm nucleus.

Since this process involves the fusion of three haploid nuclei, it is known as triple fusion.

It results in the formation of the endosperm.

Conclusion

One male gamete nucleus and two polar nuclei are involved in this process.

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Dormancy of
the zygote
after
fertilisation

Why do you think the zygote is dormant for sometime in a fertilised ovule?

Conclusion

The zygote forms when the male gamete fuses with the nucleus of the egg cell.

The zygote stays dormant for some time and waits for the endosperm to form, which develops from the primary endosperm cell produced by triple fusion.

The endosperm provides food for the growing embryo.

Once the endosperm has formed, the embryo resumes development from the zygote.

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Differentiating
floral and
seed
structure
terms

Differentiate between:

Part (a)

hypocotyl and epicotyl

Hypocotyl

1. The portion of the embryonal axis which lies below the cotyledon in a dicot embryo is known as the hypocotyl.

2. It terminates with the radicle.

Epicotyl

The portion of the embryonal axis which lies above the cotyledon in a dicot embryo is known as the epicotyl.

It terminates with the plumule.

Part (b)

coleoptile and coleorrhiza

Coleoptile**Coleorrhiza**

It is a conical protective sheath that encloses the plumule in a monocot seed.

It is an undifferentiated sheath that encloses the radicle and the root cap in a monocot seed.

Part (c)

integument and testa

Integument

It is the outermost covering of an ovule. It provides protection to it.

Testa

It is the outermost covering of a seed.

Part (d)

perisperm and pericarp

Perisperm

It is the residual nucellus which persists. It is present in some seeds such as beet and black pepper.

Pericarp

It is the ripened wall of a fruit, which develops from the wall of an ovary.

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Apple as a false fruit



Why is apple called a false fruit? Which part(s) of the flower forms the fruit?

False fruit vs true fruit

A fruit that develops from the ovary along with other accessory floral parts is called a false fruit.

A true fruit, by contrast, develops from the ovary alone and does not include the thalamus or any other floral part.

Conclusion

In an apple, the fleshy receptacle forms the main edible part.

Because the fruit includes this accessory part, apple is a false fruit.

Q

15

Emasculation technique in plant breeding

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What is meant by emasculation? When and why does a plant breeder employ this technique?

Emasculation

Emasculation is the removal of anthers from a bisexual flower without affecting its female reproductive part, the pistil.

It is used in various plant hybridization techniques.

Conclusion

Plant breeders emasculate bisexual flowers to obtain a desired plant variety by crossing a particular plant with a desired pollen grain.

The flowers are covered with a bag before they open, so that they are pollinated only by pollen grains from desirable varieties.

Later, breeders dust mature, viable, stored pollen grains onto the bagged stigma to carry out artificial pollination and obtain the desired plant variety.

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Inducing
parthenocar-
py with
growth
substances



If one can induce parthenocarpy through the application of growth substances, which fruits would you select to induce parthenocarpy and why?

Parthenocarpy

Parthenocarpy is fruit development without fertilization or seed formation.

This technique is used to produce seedless varieties of economically important fruits such as orange, lemon and watermelon.

Conclusion

Parthenocarpy is induced by applying plant growth hormones such as auxins to trigger fruit formation.

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Role of
tapetum in
pollen wall
formation



Explain the role of tapetum in the formation of pollen-grain wall.

Tapetum

The tapetum is the innermost layer of the microsporangium.

It nourishes the developing pollen grains.

Conclusion

During microsporogenesis, the cells of the tapetum produce enzymes, hormones, amino acids and other nutritious material needed for the development of pollen grains.

The tapetum also produces the exine layer of the pollen grains, which is made of sporopollenin.

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Apomixis
and its
importance



What is apomixis and what is its importance?

Apomixis

Apomixis is the mechanism of seed production without meiosis or syngamy.

It plays an important role in hybrid seed production.

Conclusion

Producing hybrid seeds by cultivation is very expensive for farmers.

Also, hybrid characters are difficult to maintain when hybrid seeds are sown, since characters segregate during meiosis.

Apomixis prevents this loss of specific characters in the hybrid, and it is also a cost-effective method for producing seeds.